

Informe de experimento: results_stress_gpt-4o-mini_gate3

Generado el 2026-09-16 11:39 por benchmarks/experiment_report.py a partir de 3 report(s) del directorio results_stress_gpt-4o-mini_gate3 (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]  
↪ low-agent/benchmarks/results_stress_gpt-4o-mini_gate3
```

- Reports leídos: 3 (report.rescored.json usado en 3; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: pfagent
- Casos: case14, case30, case57
- Requests: generadas, N=40 por caso y seed, difficulties=['stress'] (3 reports)
- Items por dificultad: stress=120
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 08:11
- Costo total: 0.2007 USD (120 items; costo medio 0.00167 USD/item)
- Items totales: 120; ok=120; con error=0; tokens totales=1262980
- Solved global: 100.0 % (120/120)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do  
You complete tasks by calling tool functions:  
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)  
2. Run AC power flow  
3. Modify the load at a bus and re-solve  
4. Disconnect or reconnect a line and re-solve  
5. Query the network status  
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)  
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk  
  
## Key rules  
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.  
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.  
- If the request is ambiguous, state the interpretation you are using.  
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.  
- After a network modification, re-solve before reporting any value.  
- Always reply in English.  
  
## Reply format  
After each analysis, reply in this structure:  
1. One-sentence summary  
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs  
3. Violations, if any  
4. Suggested next step
```

User prompt (primer item) = request_text —90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfactagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
pfagent	case1440		100.0 (40/40)	–	–	100.0 (40/40)	60.0 (24/40)	–	100.0 (40/40)	100.0 (40/40)	3.6	6629	0.0 (0/40)	–	–	0.0437
pfagent	case3040		100.0 (40/40)	0	0.00	100.0 (40/40)	60.0 (24/40)	0	100.0 (39/39)	100.0 (39/39)	3.8	132420.0 (0/39)	0.0 (0/1)	0.0 (0/1)	100.0 (1/1)	0.0836
pfagent	case5740		100.0 (40/40)	–	–	100.0 (40/40)	60.0 (24/40)	–	100.0 (40/40)	100.0 (40/40)	2.9	117040.0 (0/40)	–	–	–	0.0734

Agregado por método (media ponderada por items sobre todos los casos)

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
pfagent3	casos	120	100.0 (120/120)	0	0.00	100.0 (120/120)	60.0 (72/120)	0	100.0 (119/119)	100.0 (119/119)	3.4	105250.0 (0/119)	0.0 (0/1)	0.0 (0/1)	0.8 (1/1)	0.2007

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	stress
pfagent	100.0 (120/120)

Solved por dificultad (%, items resueltos / items)

Method	stress
pfagent	100.0 (120/120)

Tipos de error de formulación (conteo de items)

Method	ok
pfagent	120

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↳ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "", "solver_backend": "pandapower",
↳ "llm_prompt": null, "llm_response": null, ...[367 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=9.78e-09 MW) -> números retenidos
--- Ronda 2 ---
LLM: The IEEE 14-bus system was loaded, and the transformer branch between bus 7 and bus 8 was disconnected. However, the power flow
↳ analysis did not converge due to bus 8 being isolated by this topology change.

1. Summary: The network is split, and bus 8 is isolated.
2. Key numbers: Total load and gene ...[293 chars más]
--- Respuesta final ---
The IEEE 14-bus system was loaded, and the transformer branch between bus 7 and bus 8 was disconnected. However, the power flow analysis
↳ did not converge due to bus 8 being isolated by this topology change.
```

```
1. Summary: The network is split, and bus 8 is isolated.
2. Key numbers: Total load and generation values are not available due to non-convergence.
3. Violations: Bus 8 is isolated, leading to a non-converged solution.
4. Suggested next step: Consider reconnecting the line between bus 7 and bus 8 or modifying the network to restore connectivity. Would you
↳ like to reconnect the line?
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=2039;
↳ cost_usd=0.0004; wall=3.6 s
```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-stress-001-s0, case14, difficulty=stress, seed=827308000): Load the 14-bus test case (case14), disconnect the line between
bus 2 and bus 4, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line
↳ whose loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 270.1361769951196,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 2, "to_bus": 4})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.717168458565059, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5946 chars más]
gate: PASS (reasons=, mismatch=3.08e-08 MW)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "", "solver_backend": "pandapower",
↳ "llm_prompt": null, "llm_response": null, ...[367 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=8.78e-08 MW) -> números retenidos
--- Ronda 2 ---
LLM: The power flow analysis could not be completed due to network isolation caused by the disconnection of the lines.

1. **Summary**: The network is split, and the power flow did not converge.
2. **Key numbers**: Not applicable due to non-convergence.
3. **Violations**: None reported, but the network i ...[163 chars más]
--- Respuesta final ---
The power flow analysis could not be completed due to network isolation caused by the disconnection of the lines.
```

```
1. **Summary**: The network is split, and the power flow did not converge.
2. **Key numbers**: Not applicable due to non-convergence.
3. **Violations**: None reported, but the network is isolated.
4. **Suggested next step**: Consider reconnecting one of the disconnected lines to restore network connectivity and then rerun the power
↳ flow analysis.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=4648;
↳ cost_usd=0.0008; wall=4.5 s
```

Ejemplo: primer item

```
request (case14-stress-002-s0, case14, difficulty=stress, seed=1627694679): Load the 14-bus test case (case14), set the active load at bus
↳ 9 to 37.7 MW, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line whose
↳ loading is above 80%.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7}) -> disconnect_line({"from_bus": 7, "to_bus":
↳ 8}) -> run_powerflow({})
```

```

executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7}) -> disconnect_line({"from_bus": 7, "to_bus":
  ↳ 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 9, "p_mw": 37.7})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -5.20987075244386, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu ...[6276 chars más]
gate: PASS (reasons=, mismatch=7.12e-09 MW)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "", "solver_backend": "pandapower",
  ↳ "llm_prompt": null, "llm_response": null, ...[367 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.62e-08 MW) -> números retenidos
--- Ronda 3 ---
LLM: The power flow analysis could not be completed due to network isolation caused by the disconnection of the transformer branch between
  ↳ bus 7 and bus 8.

1. **Summary**: The network is split, and the power flow did not converge.
2. **Key numbers**: Not applicable due to non-convergence.
3. **Violation ...[197 chars más]
--- Respuesta final ---
The power flow analysis could not be completed due to network isolation caused by the disconnection of the transformer branch between bus 7
  ↳ and bus 8.

1. **Summary**: The network is split, and the power flow did not converge.
2. **Key numbers**: Not applicable due to non-convergence.
3. **Violations**: None reported due to non-convergence.
4. **Suggested next step**: Consider reconnecting the line between bus 7 and bus 8 to restore network connectivity and then rerun the power
  ↳ flow analysis.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=5702;
  ↳ cost_usd=0.0009; wall=4.2 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- Todos los métodos que llegan al solver (pfagent) tienen $V_MAE < 1e-4$ p.u. en todos los casos (máximo 0 en pfagent/case30).
- El gate verificó 290 resultados de flujo, 122 fallaron la verificación y se activó (retuvo números) 48 veces en 48 items.
- Ningún item terminó con **error**.
- Solved de **pfagent**: 100.0 % (120/120).
- Formulación exacta agregada sobre métodos con tools: 100.0 % (120/120).
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- De 119 items en estado de fallo, 119 lo declararon de forma segura (SFR 100.0 % (119/119)).
- 26 de 120 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- 119 items esperaban un resultado no convergido (non_converged=71, islanded=48); el método lo declaró correctamente (solved) en 100.0 % (119/119), con solved_reason declared_failure=119.
- Costo total 0.2007 USD; el método más caro fue **pfagent** con 0.2007 USD (100 %).
- Métodos con tools: 3.42 tool calls medias por item (máximo 6).

6. Archivos

Rutas absolutas bajo `results_stress_gpt-4o-mini_gate3.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
  ↳ /power-flow-agent/benchmarks/results_stress_gpt-4o-mini_gate3/case14/report.json [leído:
  ↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_stress_gpt-4o-mini_gate3/case14/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_stress_gpt-4o-mini_gate3/case14/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_stress_gpt-4o-mini_gate3/case14/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_stress_gpt-4o-mini_gate3/case14/scoreboard_per_case.json

```

